Deep Learning Approach for Earthquake Parameters Classification in Earthquake Early Warning System

Omar M. Saad, Ali G. Hafez, and M. Sami Soliman

***Abstract*— Magnitude determination of earthquakes is a mandatory step before an earthquake early warning (EEW) system sends an alarm. Beneficiary users of EEW systems depend on how far they are located from such strong events. Therefore, determining the locations of these shakes is an important issue for the tranquility of citizens as well. In light of that, this article proposes a magnitude, location, depth, and origin time categorization using earthquake Ml magnitudes between 2 and 9. The dataset used is the fore and aftershocks of the great Tohoku earthquake of March 11, 2011, recorded by three stations from the Japanese Hi-net seismic network. The proposed algorithm depends on a convolutional neural network (CNN) which has the ability to extract significant features from waveforms that enabled the classifier to reach a robust performance in the required earthquake parameters. The classification accuracies of the suggested approach for magnitude, origin time, depth, and location are 93.67%, 89.55%, 92.54%, and 89.50%, respectively.**

***Index Terms*— Deep learning, earthquake early warning (EEW) system, earthquake location, earthquake magnitude.**

I. Introduction

EARTHQUAKE early warning (EEW) system is an important tool to protect civilians and critical applications which could be in danger during strong shakes such as regular and high-speed railway trains, construction workers at height, and others. An information aspect contained in this alarm signal is the magnitude of the earthquake. The literature is very rich with articles dedicated to determine the magnitude of seismic events. Each of these articles uses one of many hypotheses explaining the nature of fault rupture [1], [2]. Olson and Allen [3] and Kanamori [4] have shown that the calculated magnitude during the first few seconds of an earthquake is proportional to the magnitude calculated from the waveform representing the whole rupture. This finding is important to the EEW systems since the size of the blind zone depends mainly on the window width, in which P-wave arrival and magnitude are estimated. Since using a long window causes the blind zone to be larger, then short-window analysis is required in order to gain more time for taking the necessary precautions before the arrival of strong waves. Therefore, many researchers have tried to use the short-window analysis such as [5], where a 1-s window is used to discriminate between the far and near sources. Although the accuracy of calculating the magnitude decreases when using a short window as discussed by Wu and Zhao [6], this less accurate magnitude is enough to send the alarm signal of the EEW systems.

A deep learning technique is currently one of the leading techniques in the field of machine learning [7] and is recently used in the field of seismology. Contrary to most of the machine learning approaches, deep learning does not need preprocessing of the input data as it deals with the raw data. It is a nonlinear technique that decomposes input data into multiple processing layers representing data with multiple levels of abstraction and has a greater ability to extract significant features from the unlabeled data [8]. Deep learning has been proposed for earthquake detection [9]–[12], seismic data inversion [13], and lithology prediction of seismic data [14].

The main output of this letter is the classification of the hyperparameters: location, magnitude, depth, and origin time of earthquakes based on convolutional neural network (CNN), using only 8-s waveform, from three stations, which end 2 s after the latest P-wave arrival time. The proposed algorithm is built to be used within the EEW system due to its fast decision, ability of being evolving, and robust performance. It uses the 1970 events with locations around the Tohoku great earthquake of March 11, 2011. The dataset used is collected from the three Japanese seismic stations N. KKWH, N. RZTH, and N. KAKH.

II. Proposed Algorithm

One of the main functions of EEW system is the determination of the hyperparameters of earthquakes within the first few seconds after the P-wave arrival time. CNN is proposed to classify the earthquake hyperparameters and extract the significant features from 8-s waveforms, from three stations, which end 2 s after the latest P-wave arrival time.

These features are fed to the softmax classifier to classify the earthquake hyperparameters. The CNN consists of three main layers: input, processing, and output layers. First, the input layer is constructed to read and store the input data as a 2-D tensor which represents the vertical component from three seismic stations. The processing layer contains several types of layers: convolutional, max pooling, and activation. Several

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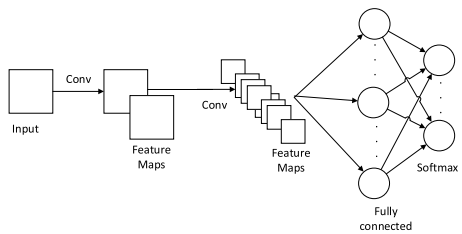

Fig. 1. CNN general topology.

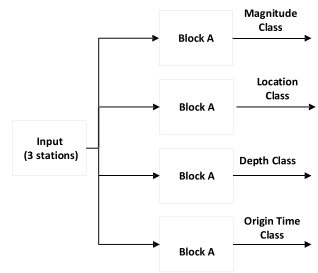

Fig. 2. Proposed CNN topology.

kernels are used in the convolutional layer to extract the significant features. Each convolutional layer extracts different significant feature maps, where each feature map output, out$[i, j]$, can be obtained as follows:

$$\text{out}[i, j] = \sum_{n=0}^{N} \sum_{m=0}^{M} (x[i+n, j+m] * k[n, m]), \tag{1}$$

where x is the input data, k is the filter coefficients (kernel), N and M are the filter order (kernel size), and i and j vary from zero to le with step size of stride, where le is the size of the input data, and the stride is the number of shifted sample for the next convolutional process. For squashing the network and nonlinearity purpose, rectified linear unit (ReLU) activation function is used, whose output $f[i, j]$ is obtained using the following formula:

$$f[i, j] = \max\ (0, \text{out}\ [i, j]). \tag{2}$$

Next, the maxpooling layer is utilized to reduce the network dimension to give the flexibility of expanding the number of feature maps and allowing the network to be deeper. The maxpooling layer output, out1$[i, j]$, is then determined by

$$\text{out1}[i, j] = \max\ (f[i+z, j+r]), \tag{3}$$

where z and r are the dimensions of the maxpooling window. The processing layer is the core of the CNN, and consecutive processing layers can be utilized to construct deep network architecture. Finally, the extracted features are fed to the softmax classifier to categorize the hyperparameters of the earthquake. The classification process is done in a supervised way, where the softmax classifier tries to match the K-mean derived clusters. The softmax output, P_c, can be obtained as follows:

$$P_c = \frac{\exp(E^T * W_j)}{\sum_{Q=1}^{LL} \exp(E^T * W_Q)}. \tag{4}$$

Here, E is the extracted features, W is the softmax classifier weights, P_c is the probability of the c class, and Q is the class number which varies from 1 to LL. The general CNN topology is shown in Fig. 1.

The proposed algorithm will function as follows: first, the P-wave arrival time of the event is estimated manually by the Japan Meteorological Agency (JMA), where this step will be made in real-time detection by one of the algorithms developed in [15]–[18]. Second, if $s_1(t)$, $s_2(t)$, and $s_3(t)$ are the seismic data from three different seismic stations, then the arrival times of the three seismic stations are compared with the latest arrival time which is set as t_0. Third, 6 s are extracted before the time t_0, and 2 s are stored after that time for the three seismic stations. The inputs for the CNN are $s_1[t_0 - 6s : t_0+2s]$, $s_2[t_0-6s : t_0+2s]$, and $s_3[t_0-6s : t_0+2s]$. Comparing the arrival times of the three stations has two benefits. First, the mutual difference between arrival times is a key point in estimating the travel times and consequently the location of the earthquake. Second, these mutual differences will decide if the event is far or near to these three stations. In case the event is near to these assigned stations (the differences are less than 6 s), then these three stations will be used in event parameters' classification. In case these differences are greater than 6 s, it means that the recorded event is far, and the other three stations should be used in classification process. In this study, as the sampling rate is 100 samples/S, and the input is a 2-D matrix with a size of 3×800 samples. The proposed algorithm has four identical architectures of CNN topologies, and each topology obtains different earthquake hyperparameters. Therefore, the input is copied four times to be fed to the four CNN topologies. In order to reach the best performance of the proposed algorithm, the number of convolutional layers, the number of feature maps, the kernel size, the dropout rate, and the type of activation function are tuned using the following procedure.

1) Fixing the network parameters and varying the number of convolutional layers.
2) Using the optimum network parameters obtained by the previous step, varying the number of feature maps in each layer.
3) Varying the kernel size using the optimum network parameters obtained by the previous steps.
4) Varying the type of activation function, using the optimum network parameters obtained by the previous steps.
5) Varying the dropout rate, using the optimum network parameters obtained by the previous steps.

In each step, the accuracy of the proposed algorithm is determined, and the highest accuracy refers to the best architecture. Hence, five convolutional layers with 16, 32, 64, 128, and 256 feature maps are enough for a robust classification performance. Each feature map has a kernel size of 3×3. The convolution operation is done by sliding this filter over the input. The stride of the convolutional filter is one sample in each step. For the output layer, the fully connected layer is used to wrap up the CNN topology. In order to train the network faster and add kind of regularization, batch normalization layer is used after each convolutional layer. The dropout layer with a rate of 0.2 is used after each pooling layer to avoid the overfitting problem. Finally, the extracted features are fed to the softmax classifier to obtain the final output. The proposed CNN topology is shown in Figs. 2 and 3.

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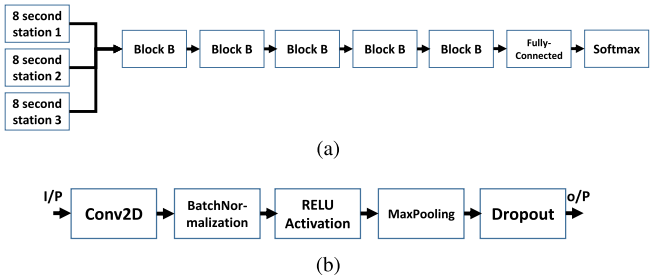

Fig. 3. (a) Architecture of Block A. (b) Architecture of Block B, note that the number of feature maps is doubled each Block B, such as the first Block B has 16 feature maps and the last Block B has 256 feature maps.

For training process, each CNN topology, Block A, is trained separately using Adam optimizer [19] with a learning rate of 0.001. The dataset is split randomly to 70% for training and 30% for testing. The number of epochs is 120, and the batch size is 16.

III. Dataset

The data from the Japanese Hi-net seismic network has been used in this letter to provide the datasets for training and testing. The earthquake catalog used in this letter is produced by the JMA [20]. The catalog is based on seismic data provided by the National Research Institute for Earth Science and Disaster Resilience, the JMA, and various institutions. Python scripts were used for creating the dataset. "WIN32" waveform data and arrival times were downloaded and converted using the HinetPy [21]. Karakuwa, N. KKWH, Rikuzentakata N. RZTH, and Kahoku N. KAKH stations are selected to construct the used dataset because they are the most used picks of phase arrivals according to the JMA bulletin [22], and accordingly, 1970 events during the period from January 1, 2010 to December 31, 2011, have been used with a total of 5910 traces. All selected events occur within the region having 141° E < Longitude < 144.5° E and 37.4° N < Latitude < 39.2° N. The events' map is shown in Fig. 4. Each dataset record consists of the waveform data segment from the vertical component for the three stations with 8 s length, 2 s after the latest P-wave arrival time, and 6 s before that arrival. Given that the sampling rate is 100 samples per second, therefore, every data record is 2400 (800 × 3) samples long, and the P phase time is extracted from the downloaded manual picked arrival times. Each dataset record is labeled with the matched label from the catalog information according to the required classifications: location, magnitude, and depth.

IV. Results

The proposed system classifies the detected events according to their locations, magnitudes, depths, and origin times. The state of the art of this letter is that multiple parameters had been obtained with acceptable accuracy. The real-time EEW system should have an array picking algorithm functioning to detect the P-wave arrival times. Once these times are declared, the CNN will be activated to calculate the required event parameters.

A. Location Classification

In traditional location algorithms, the difference between P-wave arrivals is the key factor for determining the location

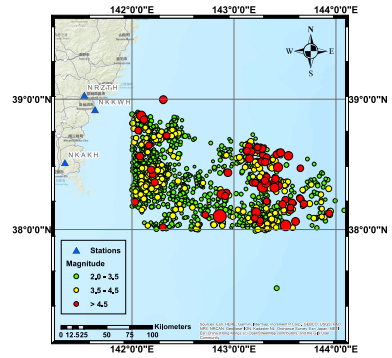

Fig. 4. Events' map for the Japanese dataset used in this article.

of an event. K-means clustering algorithm is a method that can be used for seismic zonation depending on observed seismicity instead of gathering geophysical parameters of the region of interest such as source parameters, path, and site effects. This letter proposes a novel approach for partitioning a catalog of earthquake spontaneously by using K-means on the mutual time differences between P-wave arrival times, which represents the impact of all geophysical parameters, which could be applied for any different dataset without prior knowledge about this specific region. Using the K-means algorithm, the dataset has been clustered into three classes based on these mutual time differences. These clusters are shown in Fig. 5(a). These cluster identifiers (ids) are the labels for the proposed location classifier. The accuracy of the training dataset reached 99.93%. The proposed algorithm accuracy using 30% of the dataset to test the proposed location classifier reached 94.57%. The location confusion matrix (CM_L) for the proposed algorithm is

$$\mathrm{CM}_L = \begin{bmatrix} 176 & 1 & 13 \\ 0 & 160 & 11 \\ 4 & 3 & 222 \end{bmatrix}. \tag{5}$$

For further evaluation of the proposed location classifier performance, the precision, recall, and F1-score are obtained. These parameters can be obtained as follows:

$$\begin{aligned} \text{precision(Pr)} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ \text{recall(Rc)} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\ F1-\text{score(Fsc)} &= 2 * \left(\frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}\right), \end{aligned} \tag{6}$$

where TP, FP, and FN are the true positive, false positive, and false negative, respectively. For the proposed location classifier, the precision, recall, and F1-score are 94.79%, 94.58%, and 94.60%, respectively. In order to reduce the span between the clusters, we used six clusters instead of three clusters which can significantly decrease the span between the clusters to be less than 200 km as shown in Fig. 5(b). The same procedure is done to train the proposed location classifier with 70% of the dataset and then tested with the remaining 30% of the dataset. Despite that, the proposed location classifier showed a robust classification performance. That is, the precision, recall, and F1-score have been slightly decreased to 90.67%, 90.33%, and 90.40%, respectively.

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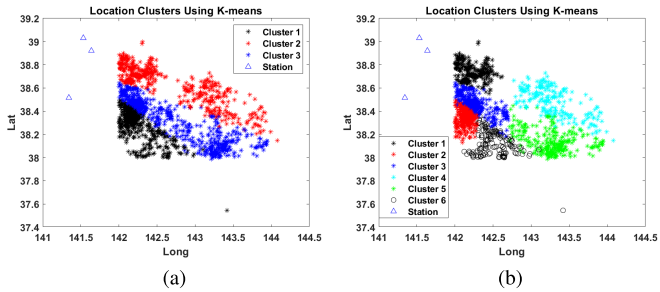
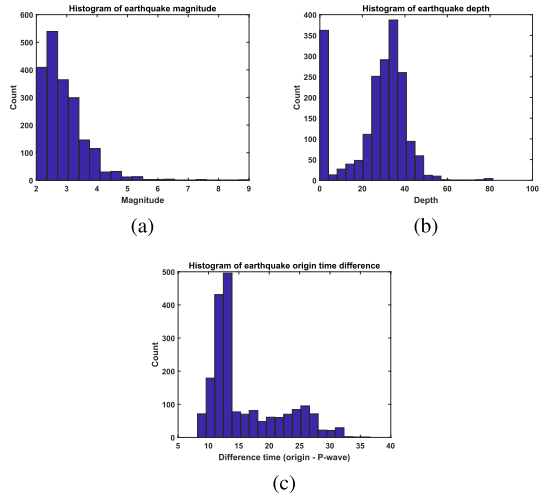


(a) (b)

Fig. 5. (a) K-mean location clustering (three clusters). (b) K-mean location clustering (six clusters).



(a) (b)

(c)

Fig. 6. (a) Histogram of dataset magnitude. (b) Histogram of dataset depth. (c) Histogram of dataset time difference between origin and arrival times.

B. Magnitude Classification

The proposed magnitude classifier classifies the events into three categories according to Ml, 2 Ml to 3.49 Ml, 3.5 Ml to 4.5 Ml, and greater than 4.5 Ml, and is tested with 30% of the dataset. The magnitude distribution of the dataset is shown in Fig. 6(a) and its magnitude confusion matrix (CM_M) is

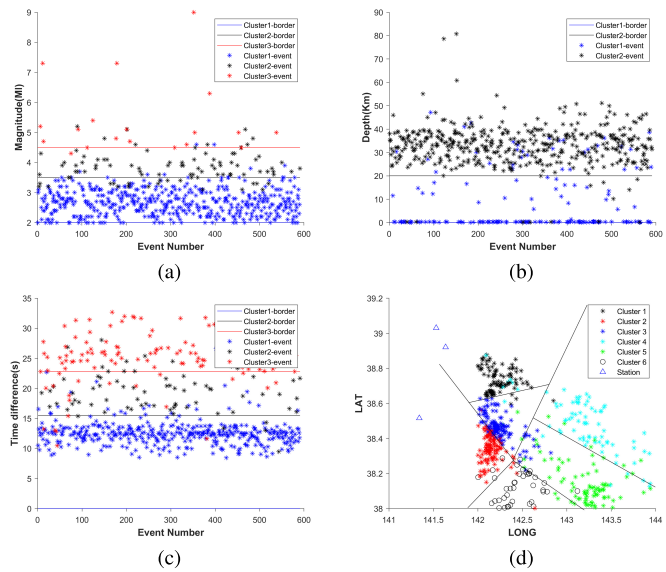
$$\mathrm{CM}_M = \begin{bmatrix} 474 & 5 & 2 \\ 22 & 60 & 8 \\ 0 & 4 & 15 \end{bmatrix}. \tag{7}$$

The training accuracy of the proposed magnitude algorithm is 98.94%, and the accuracy of the proposed magnitude classifier using dataset of testing is 93.67%. The accuracies of precision, recall, and F1-score are 92.81%, 91.77%, and 92.29%, respectively. The previous studies showed that normalizing the waveform could be used such as [23]. This step should be optimized, where the normalization process will add a step in calculations which will add a small delay in the decision of the EEW system. In this regard, the input waveforms were normalized using the maximum value which is added as an extra input for the proposed network. The classifier accuracy improved slightly, e.g., the magnitude classifier accuracy enhanced to 93.72% instead of 93.67%. Therefore, we decided to keep the algorithm without using normalization step.

C. Depth Classification

Two categories assigned to the proposed depth classifier are less than or equal to 20 km and higher than 20 km. The depth distribution of the dataset is shown in Fig. 6(b). The proposed depth classifier is trained with 70% of the dataset with an accuracy of 99.56% and tested with the remaining 30% reaching an accuracy of 92.54%. Meanwhile, the accuracy of precision, recall, and F1-score of the proposed depth classifier is 92.55%, 92.48%, and 92.52%, respectively. The depth confusion matrix (CM_D) of the proposed depth classifier is

$$\mathrm{CM}_D = \begin{bmatrix} 124 & 32 \\ 13 & 422 \end{bmatrix}. \tag{8}$$



(a) (b)

(c) (d)

Fig. 7. (a) Border of the three magnitude clusters. (b) Border of the two depth clusters. (c) Border of the three origin time clusters. (d) Border of the six location clusters. The colored star represents the event class number according to the proposed algorithm.

D. Origin Time Classification

Time difference between expected origin time and the earliest P-wave arrival from the three stations is denoted as t_d. This value is divided by the K-means into three clusters: less than 15.5 s, 15.5–22.6 s, and greater than 22.6 s as shown in the histogram of Fig. 6(c). Then, the proposed origin time classifier is trained with 70% of the dataset and tested with the rest of the data. The training accuracy reached to 99.06%, whereas the accuracy of the test dataset is 89.55%. The accuracies of precision, recall, and F1-score of the proposed origin time classifier are 91.51%, 89.51%, and 90.05%, respectively. The obtained origin time confusion matrix (CM_O) for the origin time classifier is

$$\mathrm{CM}_O = \begin{bmatrix} 366 & 23 & 3 \\ 5 & 56 & 12 \\ 6 & 14 & 107 \end{bmatrix}. \tag{9}$$

V. Analysis and Result Comparison

For each event, we plot the output magnitude, depth, origin time, and location clusters corresponding to the proposed algorithm as well as the border of each cluster as shown in Fig. 7. Accordingly, most of the miss-classified magnitude, depth, location, and origin time clusters are near to the border of

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its corresponding cluster; however, few miss-classified clusters are far away from its corresponding cluster border.

For further evaluation of the proposed algorithm performance, k-fold cross-validation has been done. The dataset is split into K partitions, $K - 1$ is used for training, whereas one is used for testing. This process is repeated K times, and each time, a different partition is used. Finally, the accuracy from all K test sets are averaged where K is set to be 10. As a result, the average test accuracies of the 10 folds are 89.85%, 91.72%, 92.49%, and 91.52% for the location, magnitude, depth, and origin time classifier, respectively. Note that, the reported accuracy for the location classier is related to the six clusters' model. Accordingly, the results demonstrate the robust performance of the proposed algorithm.

The use of deep learning EEW system is a new era in the seismology field. The long short-term memory (LSTM) approach has been proposed to classify earthquakes to near source, less than 17 km, or far source, greater than 17 km [5]. However, the proposed algorithm is more comprehensive since it classifies the earthquake location, magnitude, depth, and origin time to release the alarm in a more accurate way because not only it depends on the information of far source or near source but also on several earthquake parameters.

On the other hand, one of the robust location classifier deep learning networks is ConvNetQuake [9]. It shows the ability of deep learning to classify the earthquake location in a good manner. However, the input of ConvNetQuake is 10-s waveform which does not support EEW systems. In this letter, high accuracy could be obtained using only 8-s waveforms, from three stations, which end 2 s after the latest P-wave arrival time.

VI. Conclusion

Along this letter, the magnitude, location, depth, and origin time of earthquakes had been classified using features extracted by the CNN. This categorization can be used in an EEW system due to the fast decision and high accuracy of the proposed classifier. The performance of this algorithm achieved an accuracy of 93.67%, 89.55%, 92.54%, and 89.50% for magnitude, origin time, depth, and location classifier, respectively. It is worth to say that the obtained model will work on earthquakes coming from near locations of these events used in the selected dataset. This approach can be extended to any other area by training the model using events from the new area. This system can be implemented with reasonable dataset and computation power for any other area and nearby recording stations with high accuracy of the classifiers as had been proved for the existing case study, where the execution training time is 400 s. It is necessary to highlight that this letter did not depend on the geological parameters of this area, but it used the waveform to extract the necessary features to make its classification. The simulation has been done using a PC of Intel Processor Core i7-7700HQ CPU at 2.80 GHz, Ram of 16 Gb, Nvidia Geforce 1050 GPU, and windows-10, 64-bit Operating System.

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